

AMHERST, NS, Canada

WMO: 743905

Lat: 45.8509N

Lon: 64.2605W

Elev: 23

StdP: 101.05

Time Zone: -4.00 (NAA)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.3	-18.6	-30.9	0.2	-20.8	-27.3	0.3	-18.0	16.3	3.3	14.3	-2.0	5.8	290	0.690

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	25.5	19.5	24.1	19.0	22.8	18.3	21.0	23.9	20.0	22.5	19.2	21.6	4.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	14.6	22.6	19.0	13.8	21.6	18.3	13.2	20.7	60.8	24.1	57.5	22.5	54.6	21.6	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.1	11.5	10.3	DB	-24.7	28.8	3.2	1.3	-27.1	29.8	-29.0	30.5	-30.8	31.2	-33.1	32.2	(4)
(5)			WB	-25.3	23.0	3.1	0.9	-27.5	23.6	-29.3	24.2	-31.0	24.7	-33.3	25.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-7.6	-6.3	-1.8	3.8	9.6	14.6	18.2	18.0	14.4	8.5	3.1	-2.8	(6)	
	DBStd	9.76	6.19	5.70	4.45	3.43	3.16	3.02	1.75	2.25	3.34	3.65	4.59	5.04	(7)	
	HDD10.0	2296	547	457	366	189	46	2	0	0	4	74	213	398	(8)	
	HDD18.3	4545	806	690	624	437	271	117	24	34	123	306	457	656	(9)	
	CDD10.0	847	0	0	0	2	34	140	254	248	137	27	6	0	(10)	
	CDD18.3	53	0	0	0	0	0	4	19	23	6	0	0	0	(11)	
	CDH23.3	198	0	0	0	0	5	23	74	75	21	0	0	0	(12)	
	CDH26.7	13	0	0	0	0	0	2	5	4	2	0	0	0	(13)	
(14) Wind	WSAvg	4.7	5.0	5.0	5.0	4.9	4.6	4.4	4.4	4.1	4.2	4.7	4.9	5.0	(14)	
(15) Precipitation	PrecAvg	1212	112	94	102	95	92	89	100	89	86	109	110	130	(15)	
	PrecMax	1746	294	147	166	234	244	186	179	187	170	198	213	330	(16)	
	PrecMin	797	24	33	29	25	17	26	18	6	11	49	37	51	(17)	
	PrecStd	211	57	28	33	45	48	39	47	49	38	46	40	57	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	9.0	11.5	16.8	23.1	26.4	27.3	27.1	25.7	20.3	17.9	11.9	(19)	
		MCWB	10.1	8.3	9.9	12.1	16.6	19.8	20.4	19.5	19.7	17.5	16.1	11.0	(20)	
	2%	DB	6.7	5.3	8.6	13.4	19.7	23.6	25.3	25.3	23.1	18.0	15.5	9.1	(21)	
		MCWB	6.1	4.5	7.0	10.2	14.6	18.0	19.6	19.7	18.8	15.4	14.2	8.4	(22)	
	5%	DB	3.4	3.3	6.3	11.2	17.7	21.7	23.9	23.9	21.4	16.1	12.3	6.3	(23)	
		MCWB	2.4	2.3	4.7	7.7	13.0	16.7	18.9	19.2	17.9	13.6	11.0	5.1	(24)	
	10%	DB	0.9	1.8	4.3	9.4	15.6	20.2	22.6	22.5	19.9	14.5	9.6	3.9	(25)	
		MCWB	-0.1	0.8	2.9	6.4	11.6	16.1	18.4	18.4	16.8	12.4	8.1	2.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	8.3	10.6	13.9	17.5	21.4	22.1	22.4	21.4	18.1	16.5	11.2	(27)	
		MCDB	10.8	8.8	11.2	16.4	21.3	24.7	25.4	25.0	24.5	20.0	17.7	11.5	(28)	
	2%	WB	6.2	4.8	7.5	11.2	15.7	19.2	20.8	20.9	19.7	16.2	14.3	8.5	(29)	
		MCDB	6.9	5.2	8.4	13.1	18.8	22.0	23.8	23.6	21.8	17.8	15.5	9.1	(30)	
	5%	WB	2.6	2.5	5.1	8.6	14.1	18.0	19.8	19.9	18.7	14.2	11.3	5.3	(31)	
		MCDB	3.3	3.2	6.0	10.5	16.7	20.8	22.4	22.4	20.6	15.4	12.3	5.9	(32)	
	10%	WB	0.1	1.0	3.1	6.9	12.3	16.9	19.0	19.1	17.4	12.6	8.6	2.8	(33)	
		MCDB	0.8	1.8	4.3	9.0	14.9	19.2	21.5	21.5	19.3	14.1	9.6	3.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	8.7	7.4	7.8	9.5	9.6	8.8	8.8	9.1	8.2	6.8	7.4	(35)	
		MCDBR	13.1	9.0	10.0	11.4	13.0	11.8	11.1	10.5	10.4	10.2	9.6	9.8	(36)	
	5% WB	MCWBR	13.5	8.8	9.0	8.4	8.5	7.3	6.6	6.0	6.5	7.8	9.5	9.8	(37)	
		MCWBR	13.7	8.5	10.1	10.2	11.4	9.6	9.2	8.3	8.6	9.3	9.4	9.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.295	0.288	0.323	0.349	0.372	0.388	0.416	0.395	0.357	0.327	0.306	0.289	(40)	
		taud	2.408	2.419	2.405	2.407	2.389	2.360	2.317	2.367	2.464	2.541	2.536	2.493	(41)	
	Edn at Noon	Edn at Noon	811	898	908	910	895	879	847	853	861	843	800	786	(42)	
		Edh at Noon	71	85	100	110	116	120	124	113	94	76	63	60	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.39	2.18	3.38	4.24	4.97	5.38	5.52	4.92	3.89	2.46	1.36	1.04	(44)	
	RadStd	RadStd	0.12	0.19	0.23	0.40	0.52	0.44	0.37	0.34	0.31	0.25	0.15	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.39	N/A	N/A	N/A	N/A

Nomenclature: See separate page